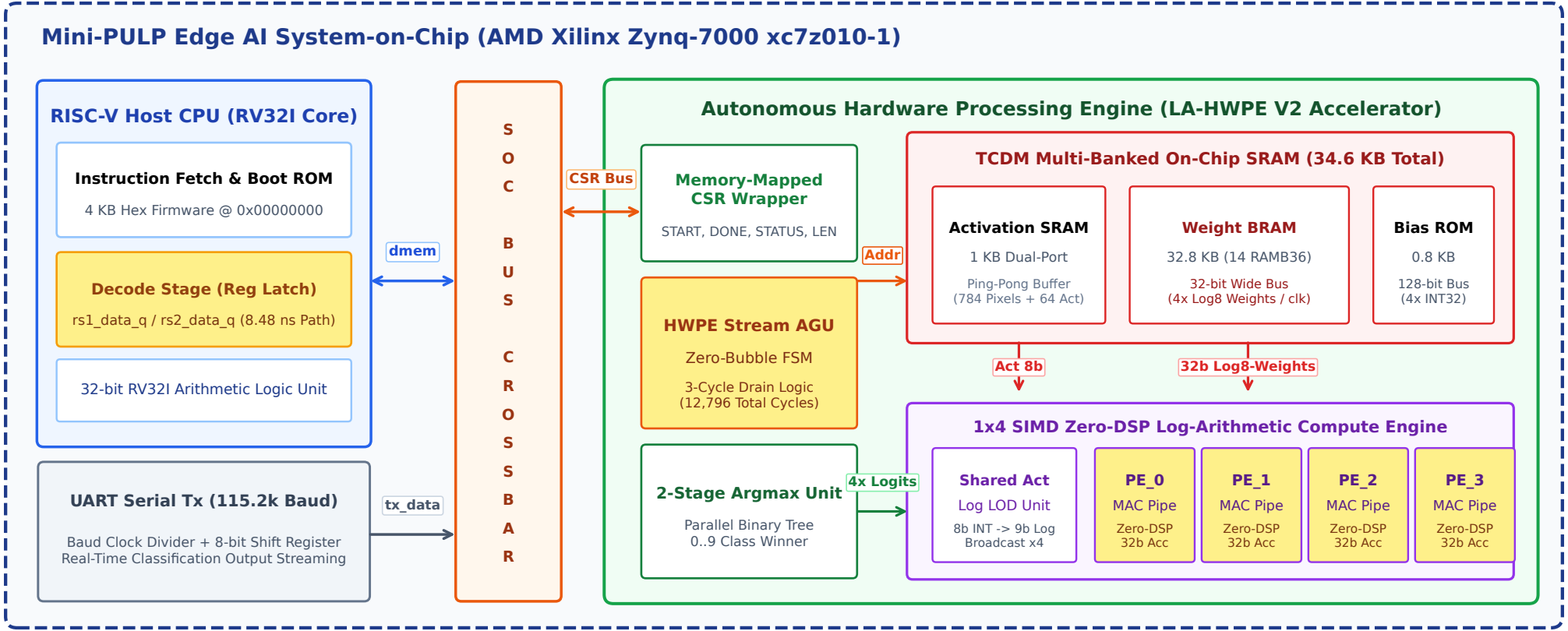
Mini-PULP Edge AI System-on-Chip (AMD Xilinx Zynq-7000 xc7z010-1)
RISC-V Host CPU (RV32I Core)
Instruction Fetch & Boot ROM
4 KB Hex Firmware @ 0x00000000
Decode Stage (Reg Latch)
rs1_data_q / rs2_data_q (8.48 ns Path)
32-bit RV32I Arithmetic Logic Unit
UART Serial Tx (115.2k Baud)
Baud Clock Divider + 8-bit Shift Register
Real-Time Classification Output Streaming
dmem
tx_data
SOC BUS CROSSBAR
CSR Bus
Autonomous Hardware Processing Engine (LA-HWPE V2 Accelerator)
Memory-Mapped CSR Wrapper
START, DONE, STATUS, LEN
HWPE Stream AGU
Zero-Bubble FSM
3-Cycle Drain Logic
(12,796 Total Cycles)
2-Stage Argmax Unit
Parallel Binary Tree
0..9 Class Winner
Addr
4x Logits
TCDM Multi-Banked On-Chip SRAM (34.6 KB Total)
Activation SRAM
1 KB Dual-Port
Ping-Pong Buffer
(784 Pixels + 64 Act)
Weight BRAM
32.8 KB (14 RAMB36)
32-bit Wide Bus
(4x Log8 Weights / clk)
Bias ROM
0.8 KB
128-bit Bus
(4x INT32)
Act 8b
32b Log8-Weights
1x4 SIMD Zero-DSP Log-Arithmetic Compute Engine
Shared Act
Log LOD Unit
8b INT -> 9b Log
Broadcast x4
PE_0
MAC Pipe
Zero-DSP
32b Acc
PE_1
MAC Pipe
Zero-DSP
32b Acc
PE_2
MAC Pipe
Zero-DSP
32b Acc
PE_3
MAC Pipe
Zero-DSP
32b Acc